

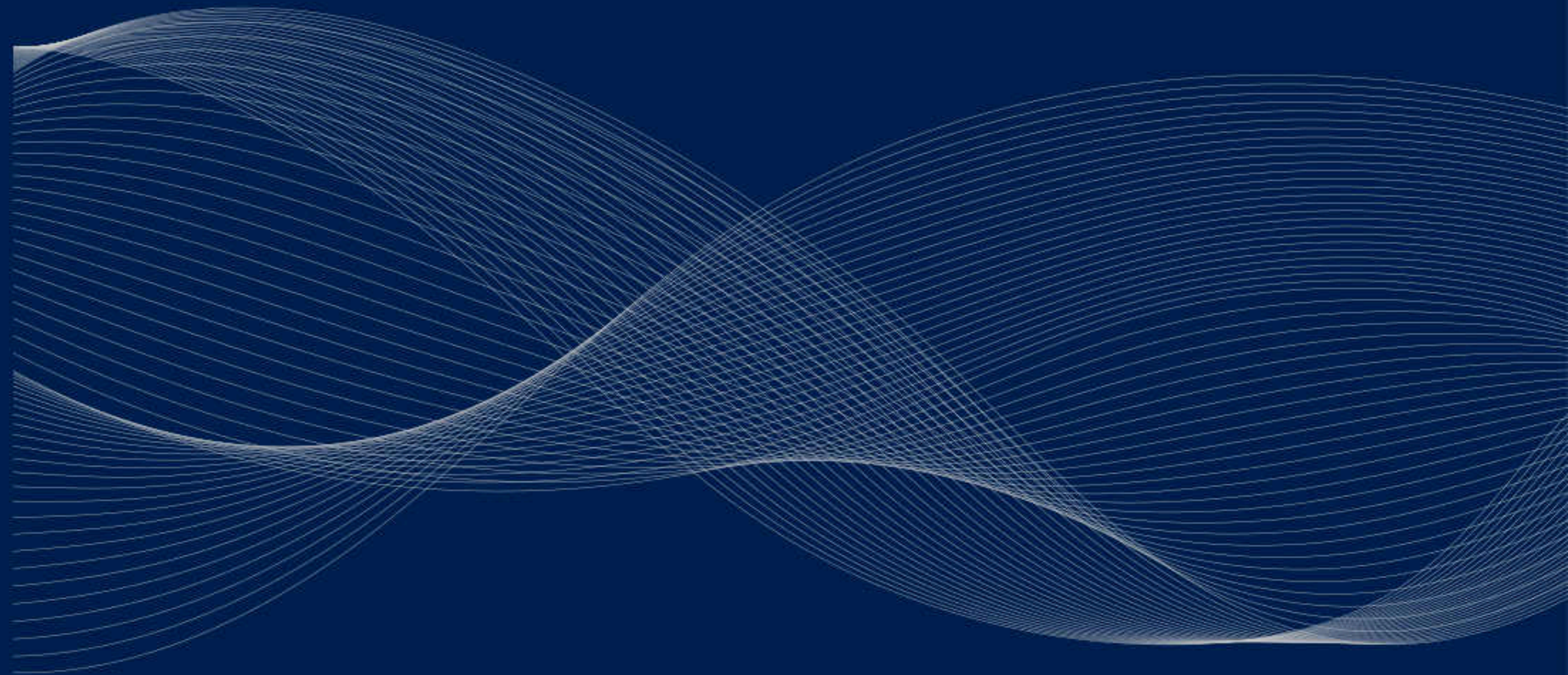


SAE-VELO

BUTT Shazil
NDIAYE Mafall
Mai 2026

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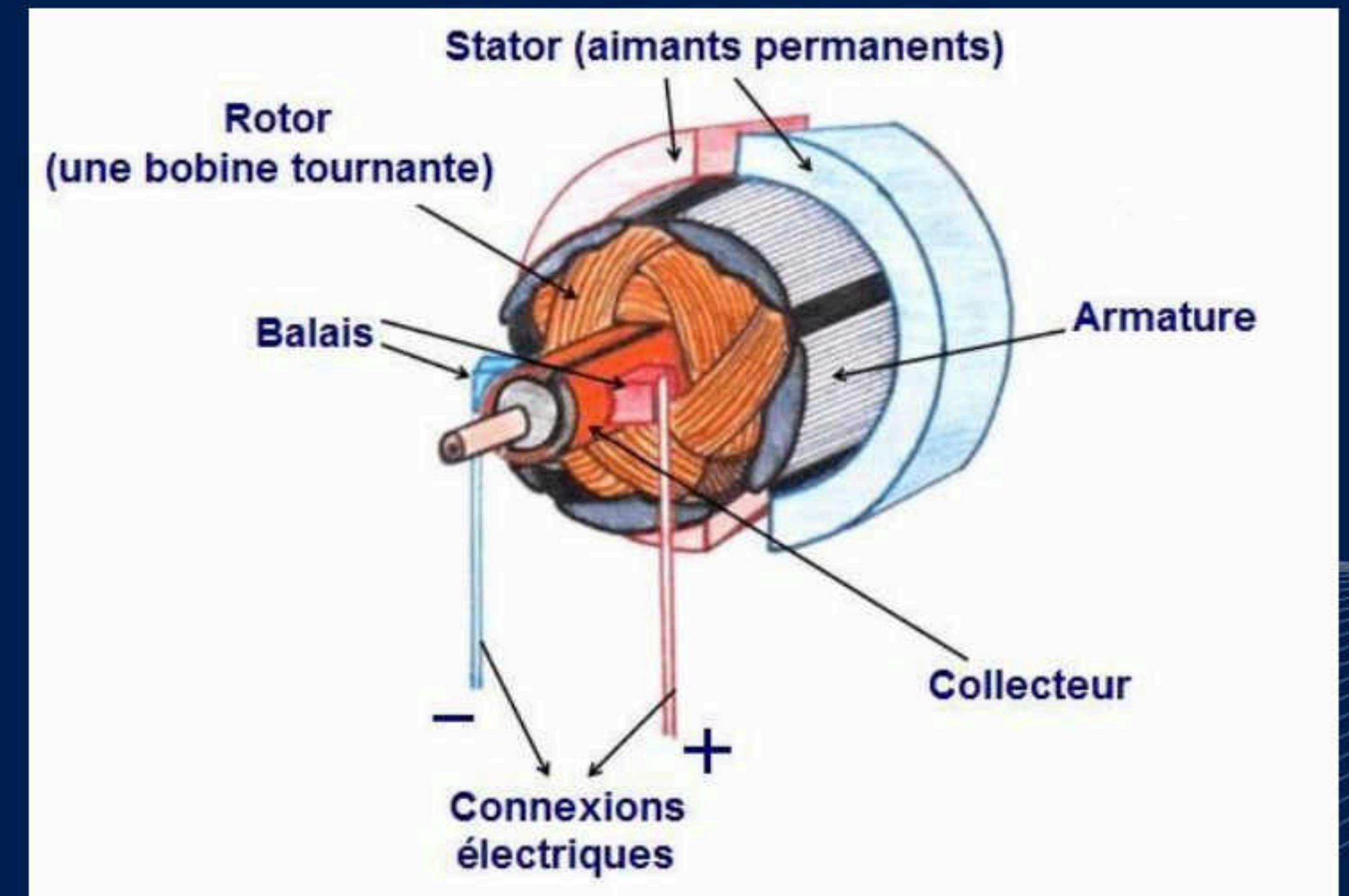
- **Présentation du Mcc**
- **Présentation du Hacheur**
- **Partie commande du Mosfet**



Le moteur à courant continu

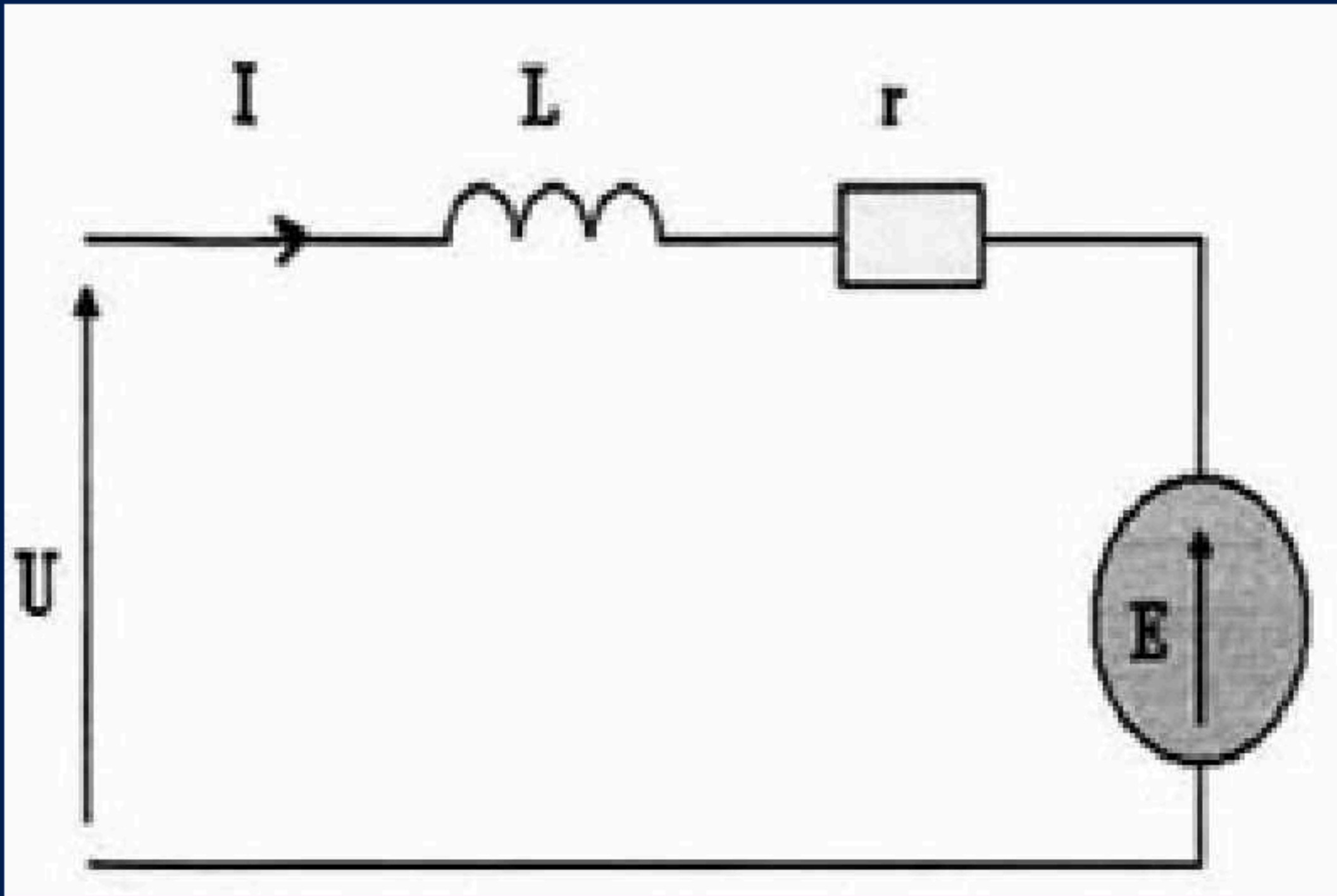
Le moteur à courant continu (MCC) est un moteur électrique qui transforme l'énergie électrique en mouvement de rotation, convertissant de l'énergie électrique en énergie mécanique.

MCC



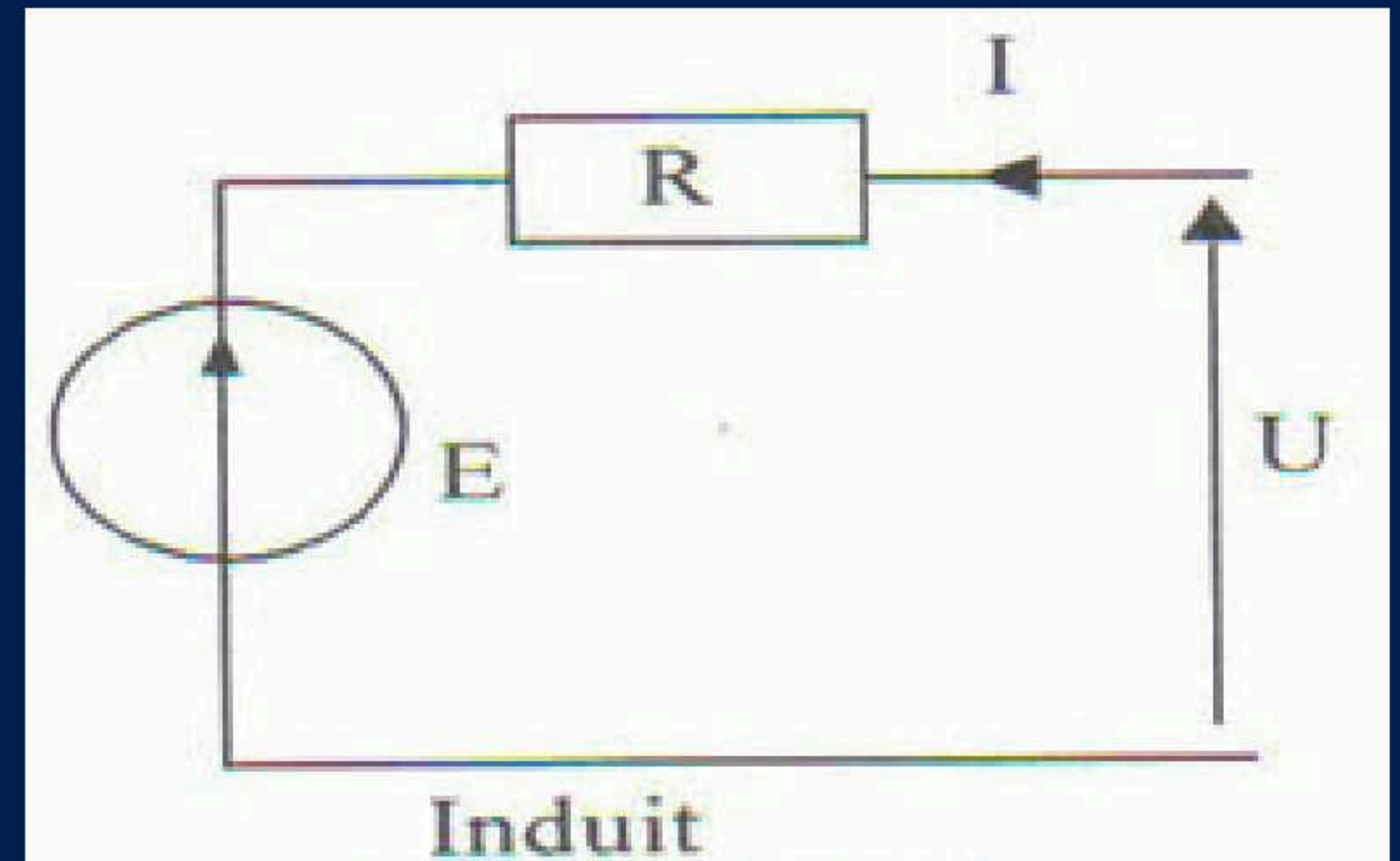
Induit

régime transitoire



$$U = R \times I + E + L \frac{di}{dt}$$

régime établi



$$U = R \times I + E$$

relation liant la vitesse de rotation angulaire du moteur et la tension U

Sachant que $E = K \times \Phi \times \Omega$

en régime établie $U = R \times I + K \times \Phi \times \Omega$

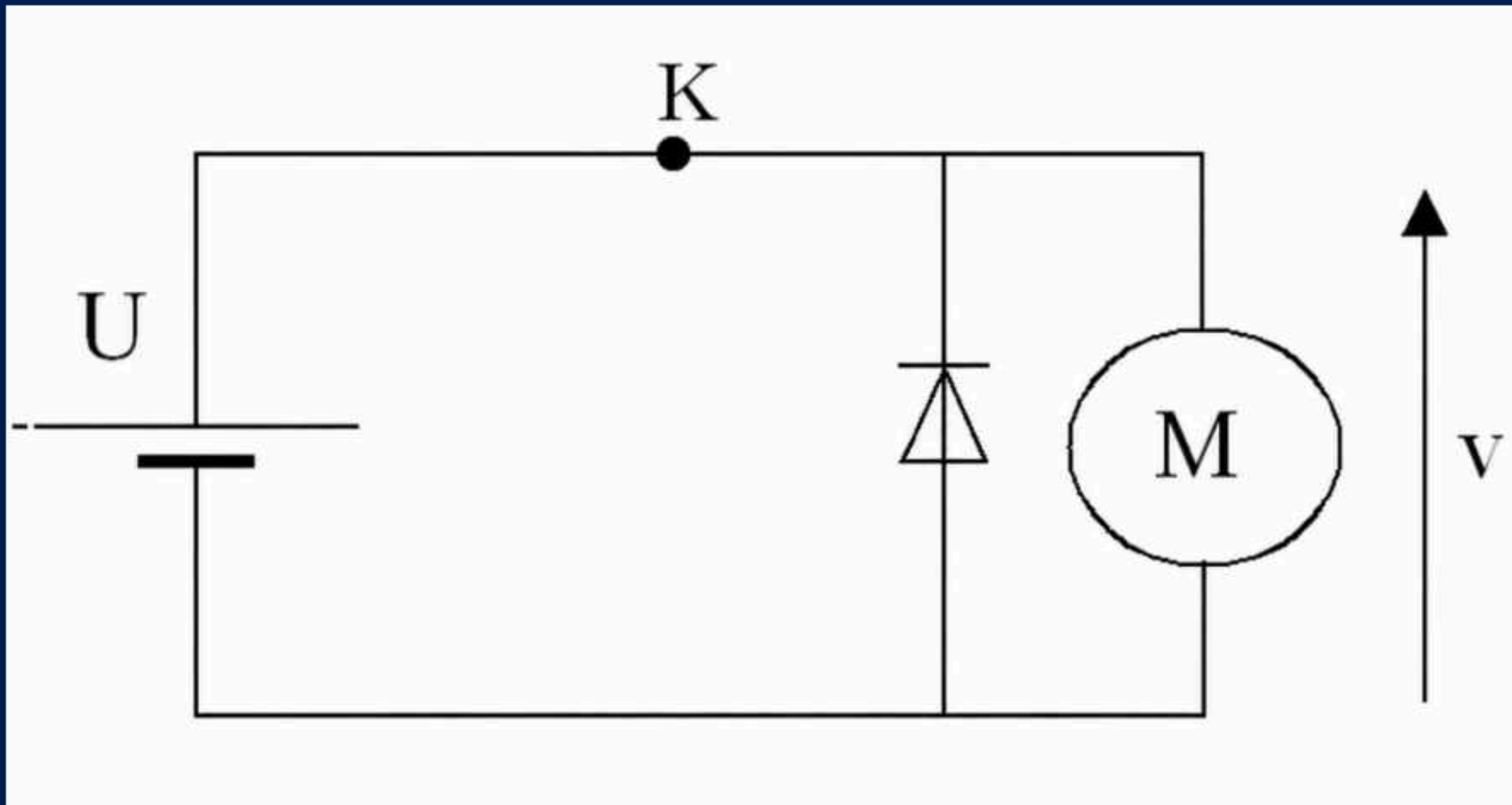
On isole Ω :

$$\Omega = (U - R \times I) / (K \times \Phi)$$

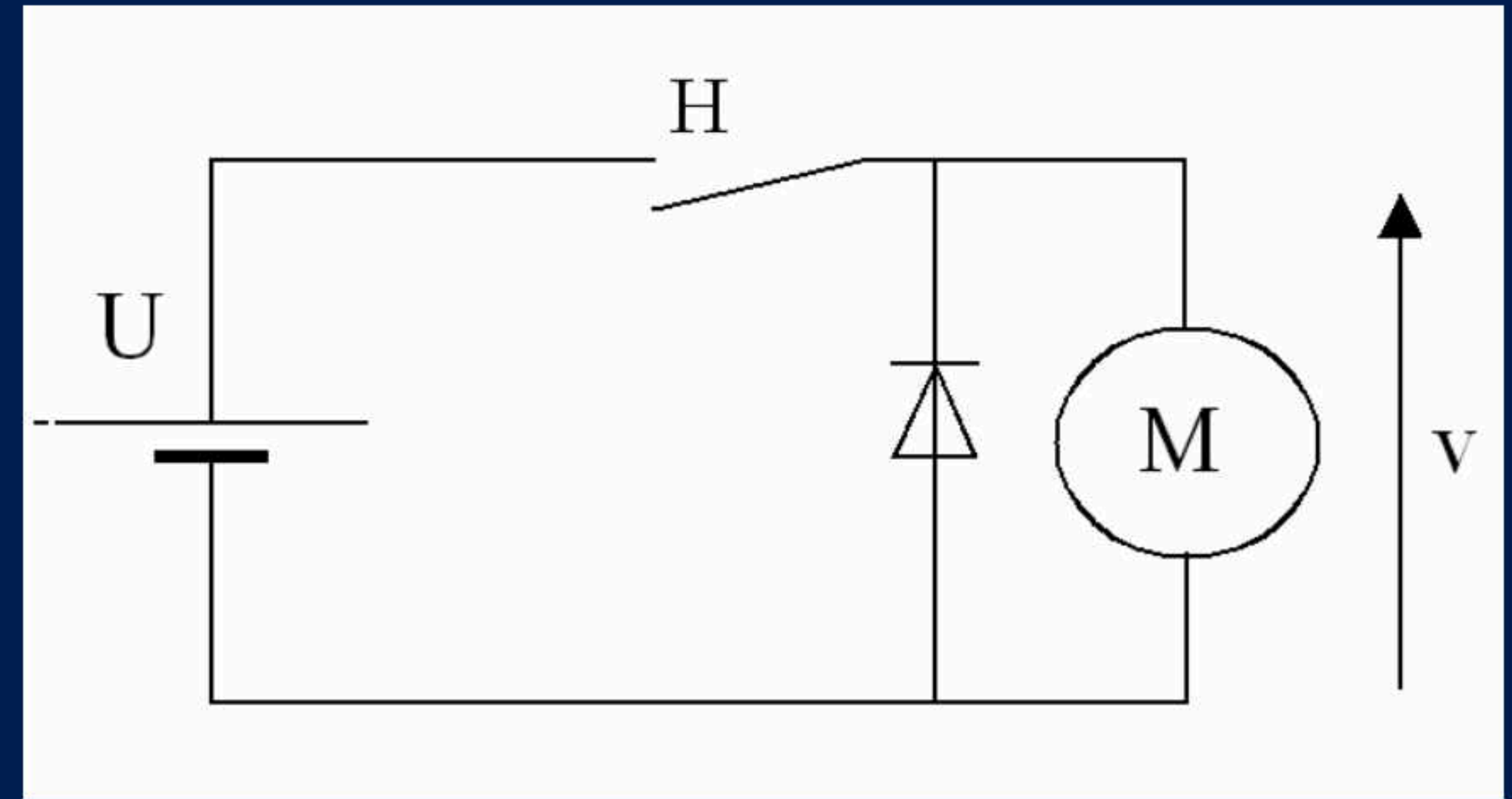
K et Φ étant des constantes en régime établi,
on peut déduire que la vitesse de rotation angulaire dépend de la tension aux bornes du moteur.

Le hacheur

Mosfet on



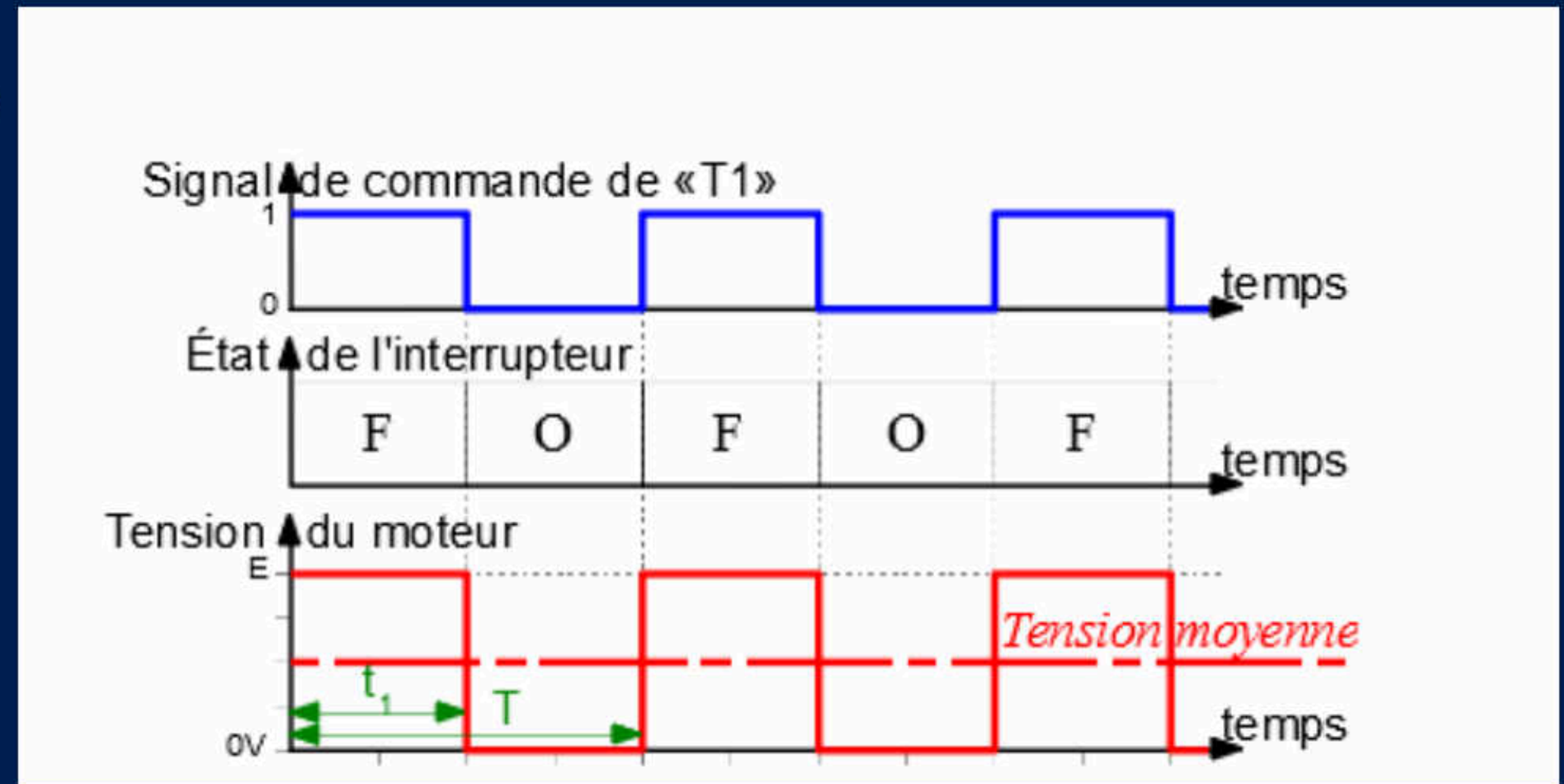
Mosfet off

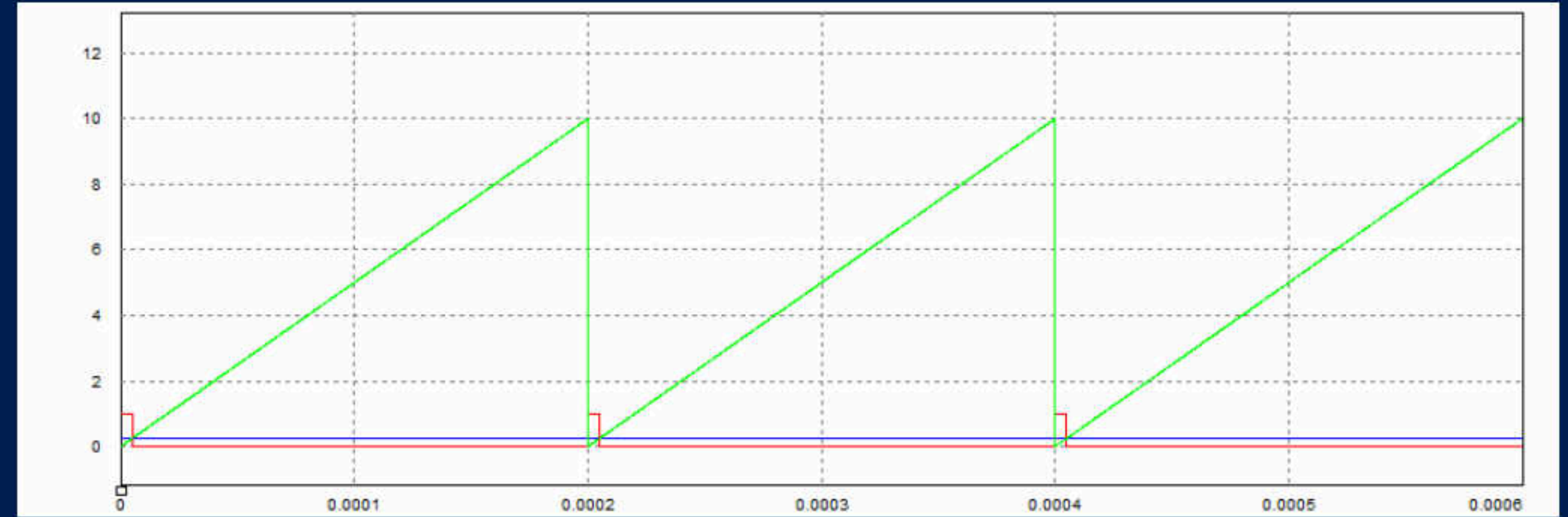


Notion de rapport cyclique

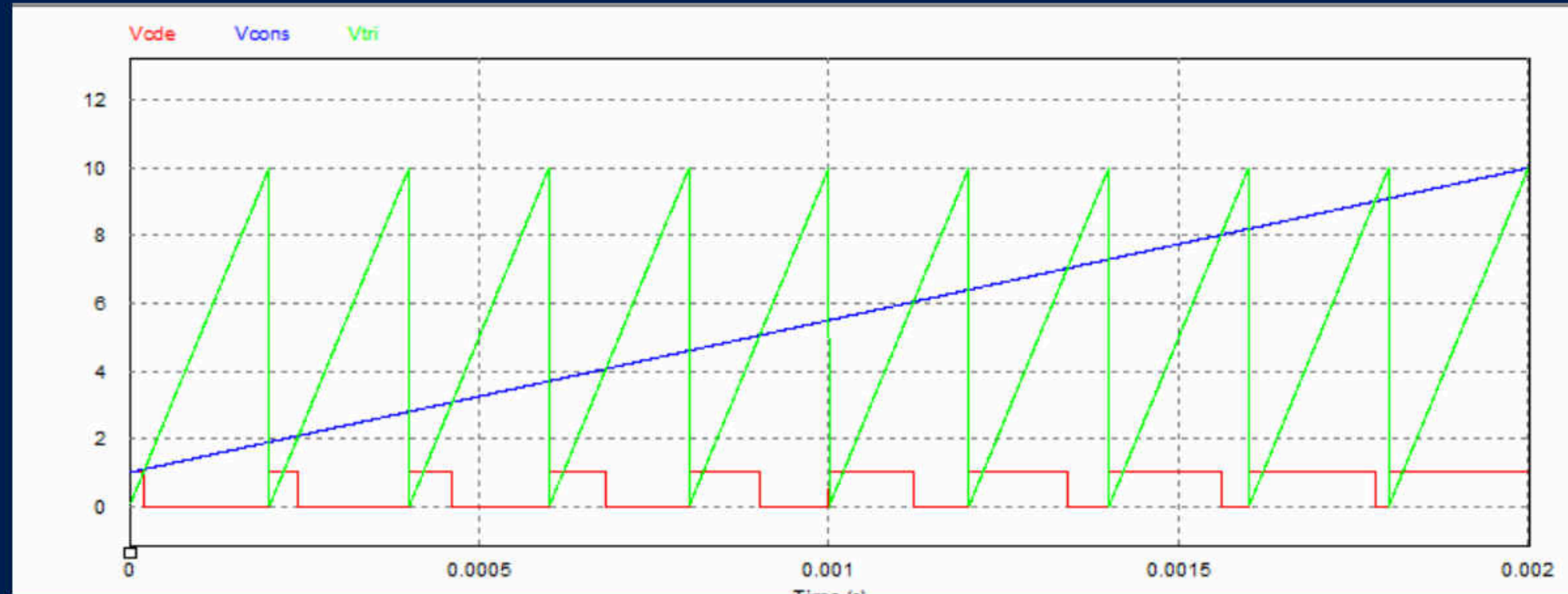
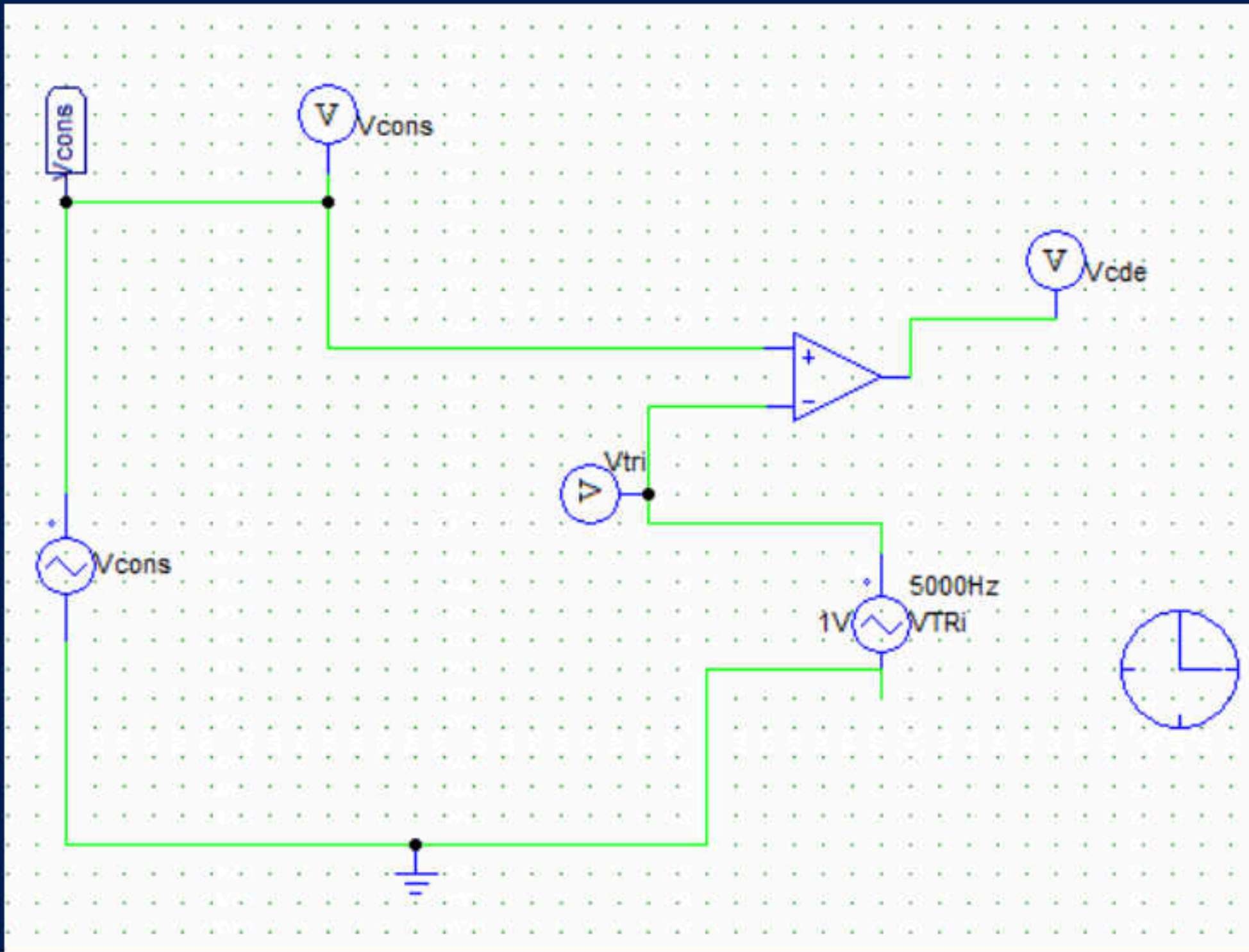
Rapport cyclique = Temps haut / Période de l'impulsion

$$U_{\text{moy}} = \text{Rapport cyclique} \times U_{\text{alim}}$$

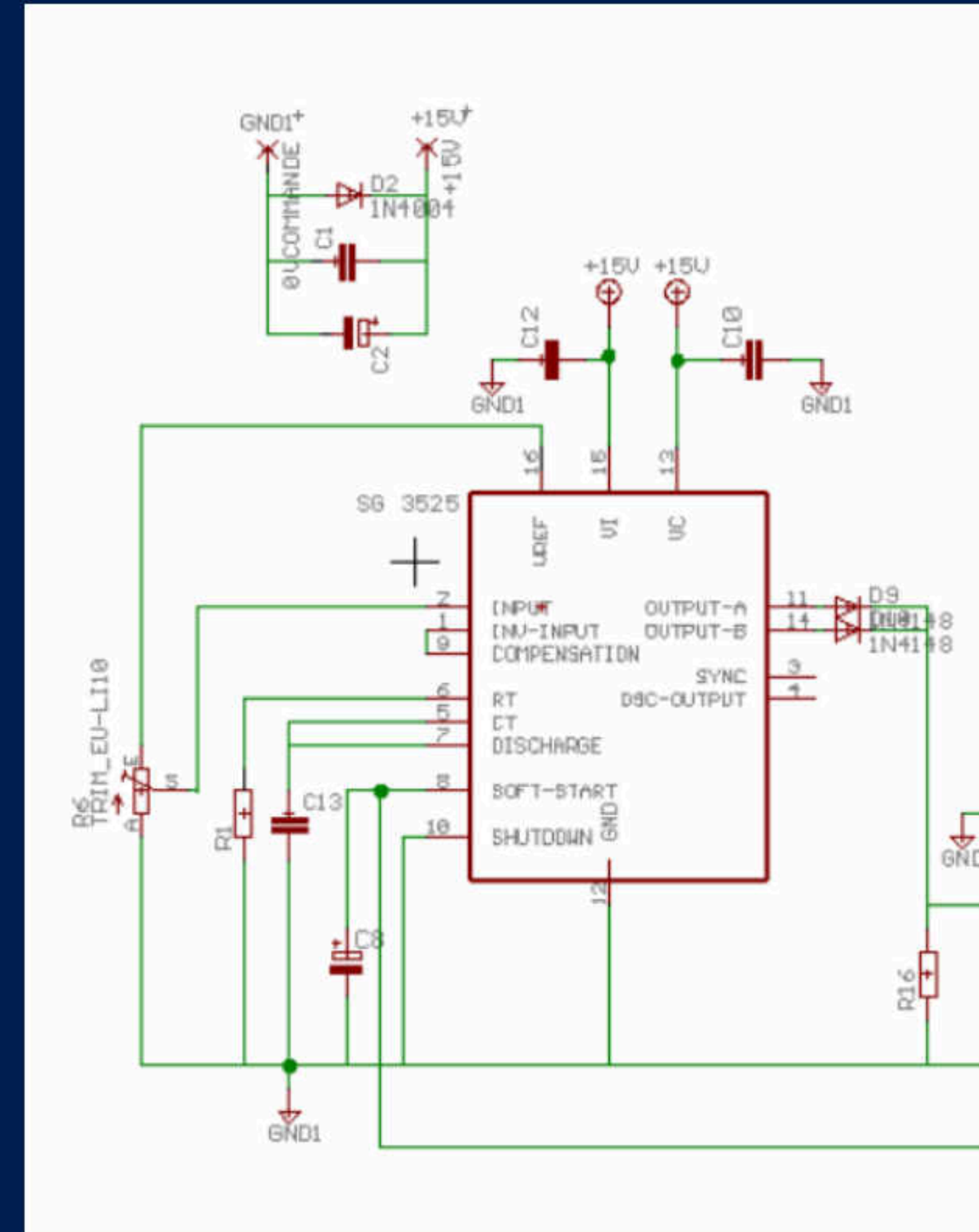
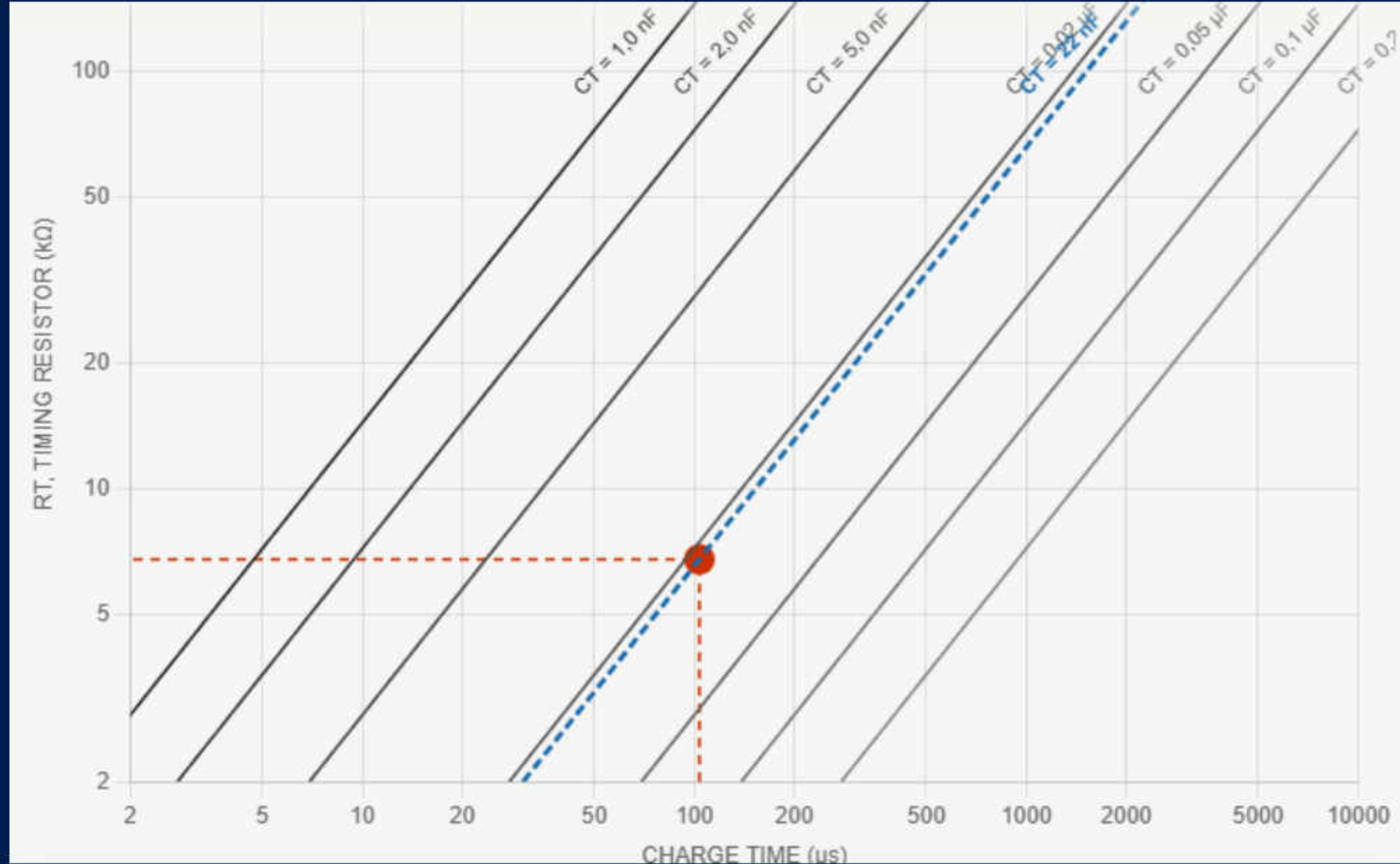




SIMULATION POUR UNE CONSIGNE QUI VARIE LENTEMENT

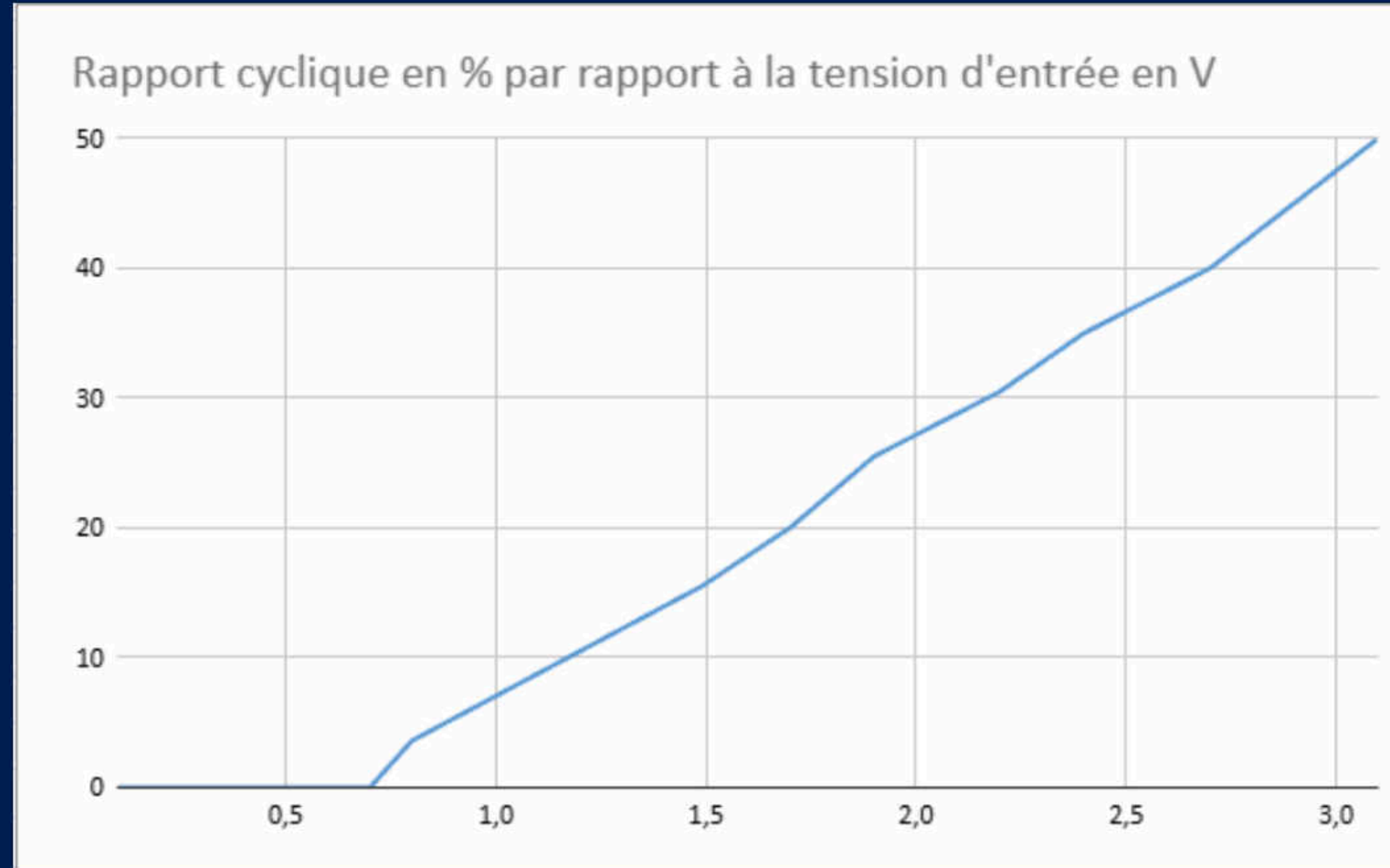


SG3525

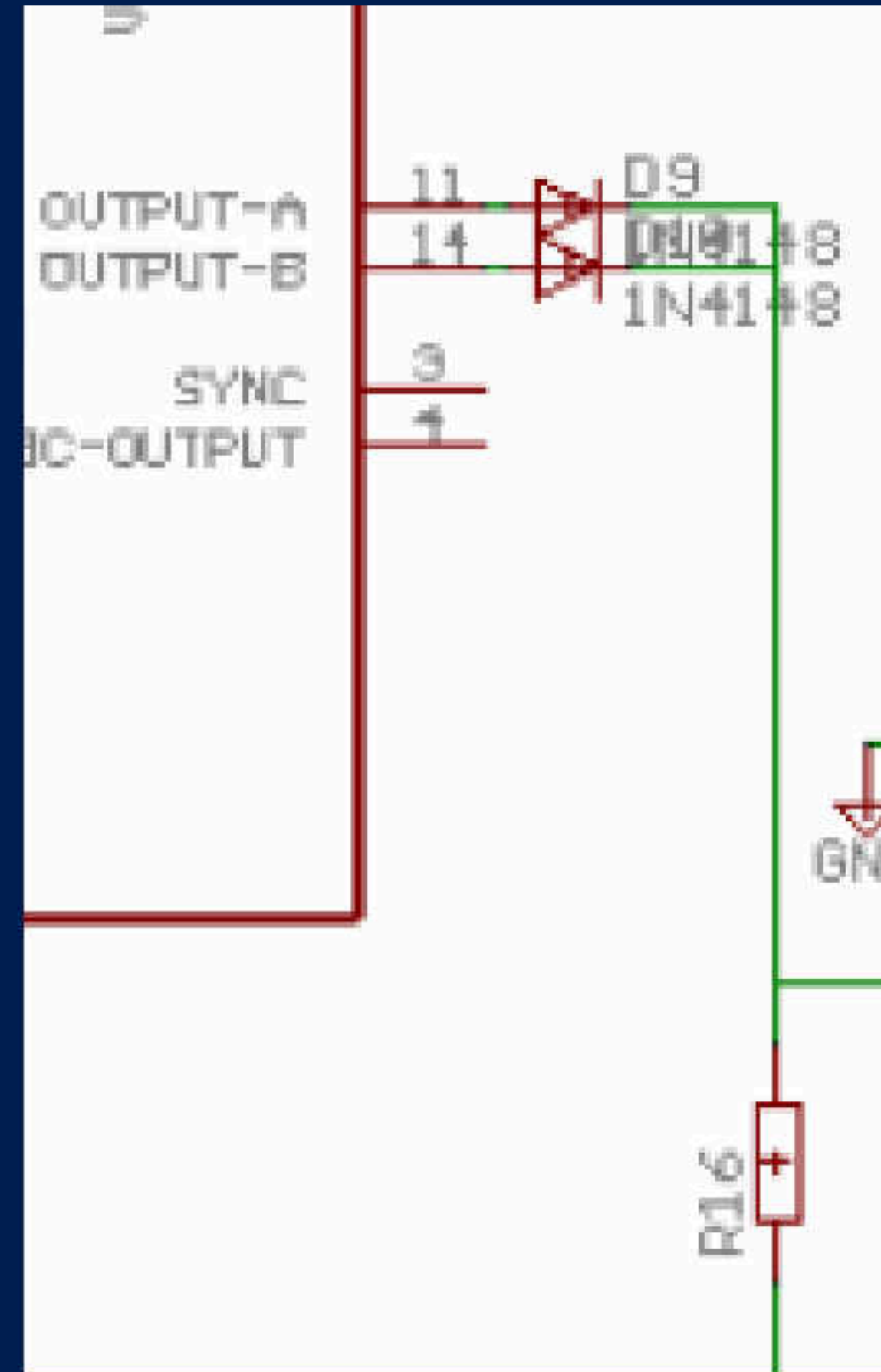


Sorties A et B

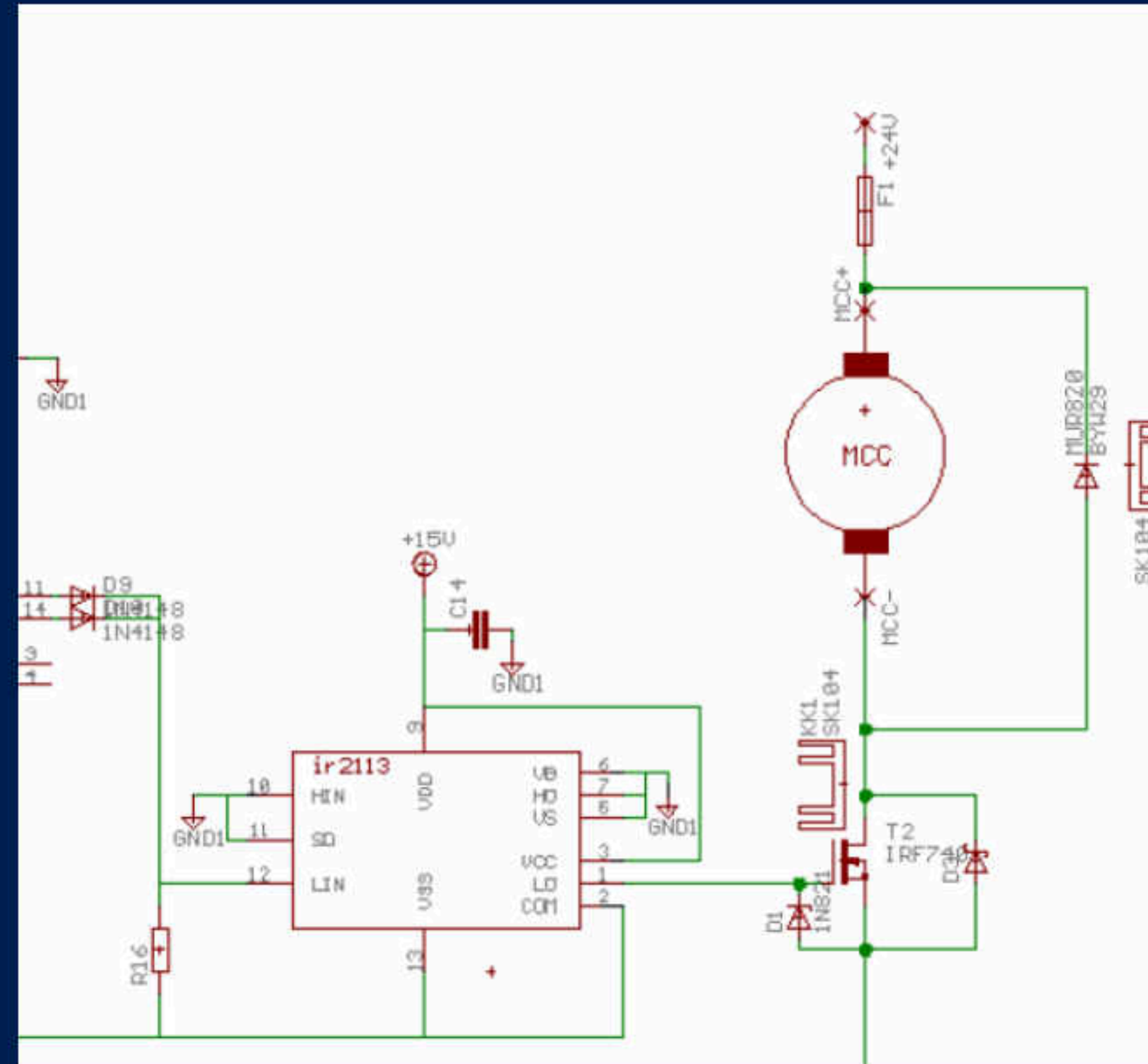




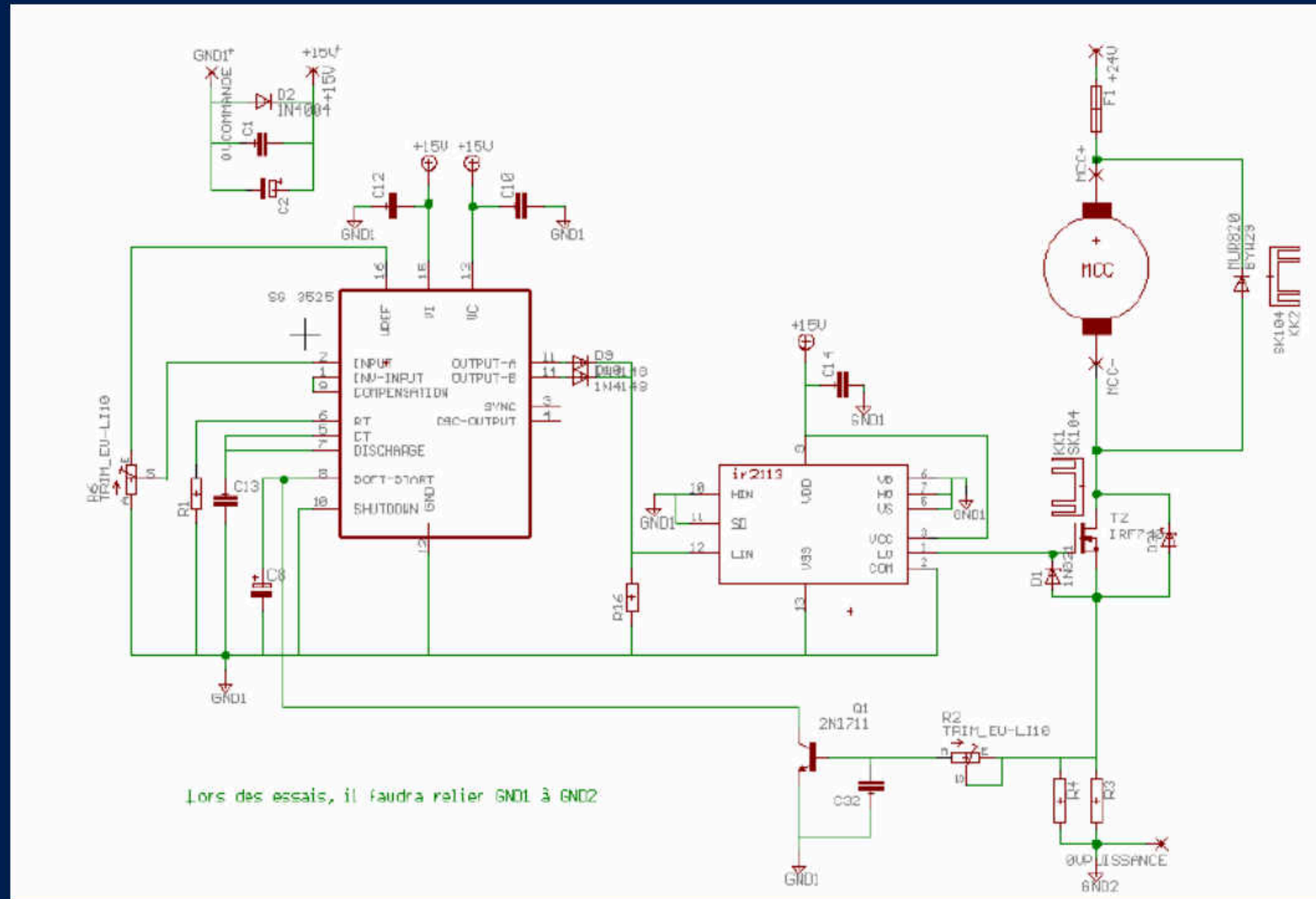
wired-OR



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Conclusion